



UNIVERSITY OF PERADENIYA

FACULTY OF SCIENCE

SEMESTER I - 2023/2024

MAT1073: Mathematics for Operations Research

End Semester Examination



Time allowed: THREE Hours

Answer All Questions

Instructions: Clearly justify each and every step of your solutions with reasoning to receive the assigned number of points. Calculators are not allowed.

Notations:

- $\mathbb{R}^n$:= Euclidean space of dimension n
 - $M_{n \times n}(\mathbb{R})$:= set of all $n \times n$ matrices with entries in $\mathbb{R}$
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1. (a) Determine whether the given set together with operations $\oplus$ and $\otimes$ is a vector space. If not, give at least one axiom that is not satisfied.

- i. The set of vectors $\{u \mid u \in \mathbb{R}\}$.

Addition: $\underline{u} \oplus \underline{v} = \underline{u} + \underline{v} + 2$

Scalar Multiplication: $c \otimes \underline{u} = c(\underline{u} + 2) - 2$, where $c \in \mathbb{R}$

- ii. The set of vectors $\{(a_1, a_2) \mid (a_1, a_2) \in \mathbb{R}^2\}$.

Addition: $(a_1, a_2) \oplus (b_1, b_2) = (a_1 + b_1 + 2, a_2 + b_2)$

Scalar Multiplication: $c \otimes (a_1, a_2) = (ca_1, a_2)$, where $c \in \mathbb{R}$

(50 marks)

- (b) Show that $W = \left\{ A \in M_{2 \times 2}(\mathbb{R}) \mid A \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} A \right\}$ is a subspace of $M_{2 \times 2}(\mathbb{R})$.

(25 marks)

- (c) Prove that $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

(25 marks)

2. (a) Express the following system of linear equations in the form $A\underline{x} = \underline{b}$ where A , $\underline{x}$ and $\underline{b}$ are matrices in appropriate dimensions.

$$\begin{aligned}x_1 + x_2 - 6x_3 &= 2 \\-3x_1 + x_2 + 2x_3 &= -10 \\x_1 - x_2 + 2x_3 &= 4\end{aligned}$$

- i. Using Gaussian elimination and back substitution, solve the system of linear equations.
- ii. Is matrix A invertible? Justify your answer.

(50 marks)

- (b) Consider the following matrix:

$$A = \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix}$$

Using the LU factorization of matrix A , solve the system of linear equations $A\underline{x} = \underline{b}$ where $\underline{b} = [-7, 5, 2]^T$.

(50 marks)

3. (a) Find the eigenvalues and the corresponding eigenvectors of the following matrix.

$$M = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

(50 marks)

- (b) Relative to a fixed origin O , the points A , B and C have position vectors $(1, -3, 2)$, $(2, 1, 1)$ and $(0, 2, 3)$ respectively.

- i. Find a vector equation for the line l that passes through the points A and C .
- ii. Find the angle $\widehat{ACB}$.
- iii. Show that the point $(-1, 7, 4)$ lies on l .

(50 marks)

4. (a) Show that the following differential equation is exact, and hence solve it:

$$(-e^{-x} + e^y) dx + (xe^y + e^y) dy = 0$$

(20 marks)

- (b) Find the solution for the following non-homogeneous equation

$$\frac{dy}{dx} = \frac{y-2}{x+y-5}$$

by using the substitutions $x = X + 3$ and $y = Y + 2$. **(20 marks)**

- (c) Using $z = y^2$ convert the following non-linear differential equation to a linear differential equation and solve it.

$$x^2 y \frac{dy}{dx} - xy^2 = 1$$

(20 marks)

- (d) Find the general solution of each of the following differential equations.

i. $D^2(D^2 + 4)y = 80x^3$

ii. $(D^2 + 4D + 4)y = e^{-2x} \sec^2 x$, where $D = \frac{d}{dx}$

(40 marks)

.END.